Supercomputers are powerful machines that are capable of computing large data sets and problems. In this report, we will look at two super computers, Eagle (developed by Microsoft Azure) and Frontier (manufactured by Hewlett Packard Enterprise). Specifically, we will explore the different architecture and how their hardware is created. Then, we will compare their computation capability, which measures how quickly they can perform calculations. We will also discuss how they use parallel multiple thread computing ability, energy consumption for operation, and also applications for artificial intelligence, and research purposes.

Firstly, we will discuss the hardware components of Eagle. It uses an Intel Xeon Platinum 8480C processor and NVIDIA H100 GPUs. The CPU is a component responsible for executing raw instructions given by the operating system. According to Top500 Eagle, it uses 2,073,600 cores and an Nvidia Infiniband NDR, a high-speed network that allows different parts of the supercomputer to communicate quickly. Eagle uses Microsoft's NDv5 computing architecture, which is closely related to the **ND-H100-v5 virtual machine series.** According to Microsoft "ND-H100-v5 sizes series" documentation, each ND-H100-v5 virtual machine has eight NVIDIA H100 GPUs, 96 vCPUs, and 1,900 GiB of memory. The virtual machine also includes 1,024 GiB of local storage and supports up to 32 remote storage disks. These large amounts of storage help Eagle's ability to build compute performance for large scale applications. Microsoft can process and compute many problems at an extremely fast pace.

Next, we will discuss the Eagle's power computational performance. According to Top500 Eagle, Eagle has a Linpack Performance (Rmax) capability of 561.20 PFop/s, theoretical Peak (Rpeak) 846.84 PFlop/s, and a Nmax of 11,796,480. The Linpack Performance (Rmax) measures Eagle's capability during a standard performance test. The Theoretical Peak Performance (Rpeak) represents maximum compute under ideal environments. These numbers show that even Eagle is capable of processing large amounts of information. According to Microsoft "What runs GPT-4o and Microsoft Copilot?", Eagle super computer uses about 14,400 NVIDIA H100 GPUs. Microsoft adds that Eagle can support training of large AI models such as OpenAI's GPT-3 model and Open-Source Heavy weights. The models need strong hardware to be able to train, fine-tune, and host massive amounts of data and deep reasoning abilities.

Thirdly, we will now analyze Eagle' parallel computing ability, its biggest advantage. It uses an Intel Xeon Platinum 8480C processor, each containing 48 physical CPU cores. Instead of a single processor, the Eagle supercomputer divides the work across many CPU cores. According to Microsoft's 'Annual Roundup of AI infrastructure Breakthroughs for 2023', Eagle uses 14,400 NVIDIA H100 GPUs connected through high-speed InfiniBand networking. If Eagle needs to solve a complicated scientific problem, it can divide the work into smaller tasks and distribute them across multiple processors and GPUs. This allows thousands of processors to work together, making it useful for calculations. In addition, the virtual machine architecture

(**ND-H100-v5 virtual**) supports network connections up to 400 gigabits per second for each GPU.

Next, we will discuss Eagle's power requirements. Eagle requires a large amount of electricity because of thousands of compute hardware and NVIDIA H100 GPUs, which can. Top500 does not publicly report the exact power consumption. These processors often produce a lot of heat and need an intricate cooling system to regulate its hardware.

After discussing Eagle's architecture, performance, parallelism, and power requirements, we will now examine another powerful supercomputer called Frontier. Similar to Eagle, Frontier is made up of many powerful CPUs and GPUs. Firstly, we will discuss its architecture and hardware. According to Top500 Frontier, Frontier uses AMD optimized 3rd Generation EPYC 64C 2GHz processor. It has around 9,066,176 cores running at 2GHz. The interconnect uses the Slingshot-11. Compared with the other super computers, Eagle has approximately 2 million cores, while Frontier has around 9 million total processing cores. This shows that Frontier has a significant number of processing cores, allowing it to perform many calculations simultaneously. Another difference is the GPU: Eagle uses NVIDIA H100 GPUs while Frontier uses AMD Instinct MI250X GPUs. Nvidia H100 offers a better theoretical performance reach up to 67 TFLOPS (NVIDIA H100 Official Specification) vs MI250X's 47.9 TFLOPS (AMD Instinct Official Specification). Thus, Eagle has a slight edge in GPU computation ability, though both GPU hardware are in a similar range.

Next, we will discuss Frontier's computation capacity and analyze it with Eagle's performance. From the Top500 Frontier, Frontier has a Linpack Performance (Rmax) of 1353 PFlop/s, a Theoretical Peak Performance (Rpeak) of 2,055.72 PFlop/s, and an Nmax of 24,837,120. Compared to Eagle, Frontier has significantly greater computational performance on the Linpack benchmark. According to TOP500, Eagle has a Linpack Performance of 561.20 PFlop/s, while Frontier achieves 1,353.00 PFlop/s. This means Frontier performs almost double in speed with Eagle.

According to Oak Ridge National Laboratory "Frontier Super Computer Hits New Highs in Third Year of Exascale", the report also confirms the Top500 findings for Frontier. It states that with a score of 1353 PFlop/s, it is the "64 bit international scientific standard for computation accuracy" (Oak Ridge). This level of performance makes Frontier useful for solving complicated scientific problems that require enormous amounts of computing power. For example, researchers can use Frontier to simulate weather patterns, study molecules, and perform complex physics calculations.

Next, we will discuss Frontier's forms of parallelism and compare them with Eagle. Similar to Eagle, Frontier uses CPU parallelism, GPU parallelism, and distributed parallelism to perform

multiple calculations at the same time. However, two supercomputers use different hardware to achieve this. According to Oak Ridge National Laboratory "Frontier Super Computer Hits New Highs in Third Year of Exascale", Frontier has 9,408 AMD EPYC CPUs and 37,632 AMD Instinct MI250X GPUs. Each computing node contains one AMD EPYC CPU and four AMD GPUs. These components work together by dividing complicated problems into smaller tasks that can be completed simultaneously. In comparison, Eagle uses Intel Xeon CPUs and 14,400 NVIDIA H100 GPUs to perform calculations in parallel, according to Microsoft. Frontier uses HPE Slingshot-11 networking, while Eagle uses NVIDIA InfiniBand NDR. Both technologies allow different computing units to exchange information quickly, which helps improve efficiency of parallel computing.

Overall, Frontier is mainly designed for large scientific simulations and research while Eagle focuses more on artificial intelligence machine learning. Overall, both supercomputers use similar forms of parallelism but their different hardware networking technologies allow them to handle large computing tasks in different ways.

In conclusion, Eagle and Frontier are two powerful supercomputers that utilize modern processors like CPUs, GPUs, and high-speed networks to perform calculations. The supercomputers selected different CPU choices and thus have different performances + applications. Eagle uses Intel Xeon processor and NVIDIA H100 GPUs, but Frontier uses AMD EPYC processor. In addition, Frontier has a fast Linpack performance benchmark score relative to Eagle's performance. Finally, their application is also different. Eagle focuses more on artificial intelligence while Frontier is focused on scientific research. Both supercomputers highlight the need for large scale computers given the digital age that we live in. Researchers and many companies need supercomputers to be able to help solve problems, process large amounts of information, and develop new technologies.

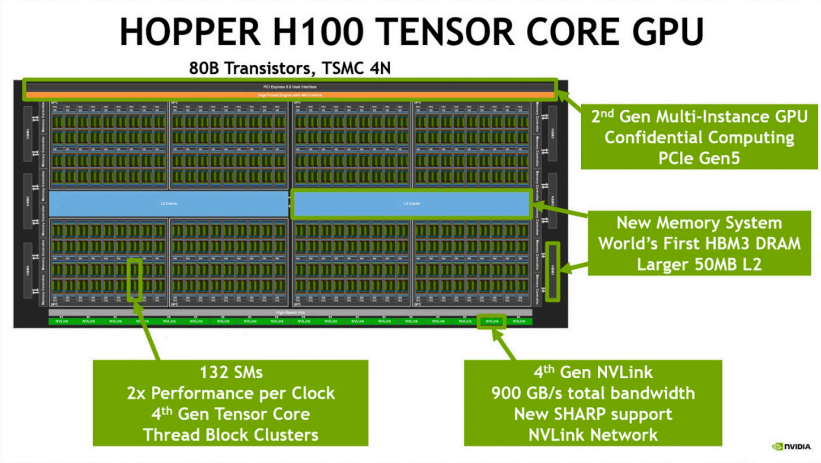


Figure 1: Architecture for NVIDIA Hopper H100 GPU



Figure 2: Frontier SuperComputer



Figure 3: Eagle Supercomputer

EAGLE - MICROSOFT NDV5, XEON PLATINUM 8480C 48C 2GHZ, NVIDIA H100, NVIDIA INFINIBAND NDR

Site:	Microsoft Azure
Manufacturer:	Microsoft Azure
Cores:	2,073,600
Processor:	Xeon Platinum 8480C 48C 2GHz
Interconnect:	NVIDIA Infiniband NDR
Installation Year:	2023
Performance	
Linpack Performance (Rmax)	561.20 PFlop/s
Theoretical Peak (Rpeak)	846.84 PFlop/s
Nmax	11,796,480
Software	
Operating System:	Ubuntu 22.04
Compiler:	NVIDIA NVCC
Math Library:	NVIDIA CUBLAS 12.2
MPI:	NVIDIA HPCX 2.16

Figure 4: Top 500 Eagle Supercomputer Stats

FRONTIER - HPE CRAY EX235A, AMD OPTIMIZED 3RD GENERATION EPYC 64C 2GHZ, AMD INSTINCT MI250X, SLINGSHOT-11, HPE CRAY OS

Site:	DOE/SC/Oak Ridge National Laboratory
System URL:	https://www.olcf.ornl.gov/frontier/
Manufacturer:	HPE
Cores:	9,066,176
Processor:	AMD Optimized 3rd Generation EPYC 64C 2GHz
Interconnect:	Slingshot-11
Installation Year:	2021
Performance	
Linpack Performance (Rmax)	1,353.00 PFlop/s
Theoretical Peak (Rpeak)	2,055.72 PFlop/s
Nmax	24,837,120
HPCG	14,054.0 TFlop/s
Power Consumption	
Power:	24,607.00 kW
Power Measurement Level:	2
Software	
Operating System:	HPE Cray OS
Compiler:	gcc/12.3.0 and hipcc
Math Library:	rocm/5.7.1 and mkl_2019
MPI:	cray-mpich/8.1.30

Figure 5: Top 500 Frontier Supercomputer Stats

Sources:

TOP500 – Eagle Supercomputer
Microsoft – AI Infrastructure Breakthroughs for 2023
Microsoft – ND-H100-v5 Documentation
TOP500- Frontier SuperComputer
Oak Ridge National Laboratory: Frontier Super Computer Hits new High in Third Year of Exascale
NVIDIA H100 Official Specification